HEALTHCARE DATA ANALYTICS

Patient Visits, Hospital Operations & Care Indicators

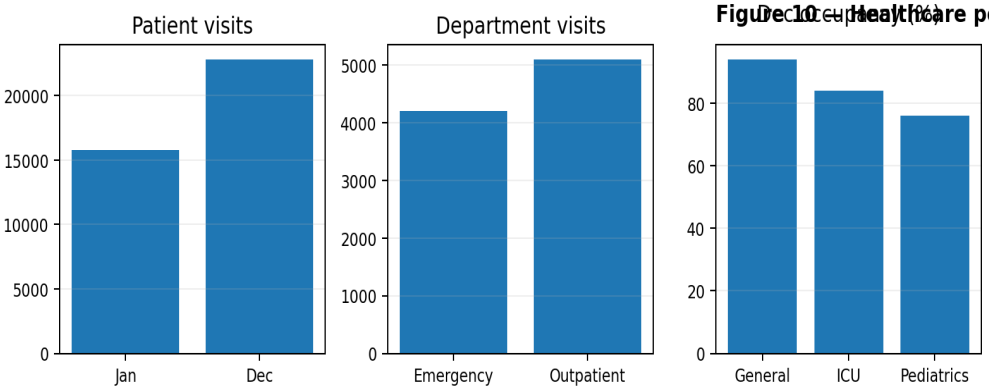


Figure 10 – Healthcare performance snapshot

VISUAL DATASET REPORT

12-month illustrative snapshot · 6 departments · 10 charts · Ordered tables

Prepared October 2026
Descriptive analytics · Simulated dataset

Important: This report contains fictional sample data for coursework and visualization practice. It is not clinical evidence and must not be used for medical decisions.

Contents & Methodology

I. Patient volume & department activity — Figures 1–3
II. Case mix, demographics & admissions — Figures 4–6
III. Treatment, capacity & readmissions — Figures 7–9
IV. Summary metrics, findings & recommendations — Figure 10
V. Conclusion and data limitations

Purpose

This report demonstrates how healthcare operations data can be summarized to understand patient volume, department workload, waiting times, case mix, admissions, treatment categories, bed occupancy and readmission trends.

Dataset design

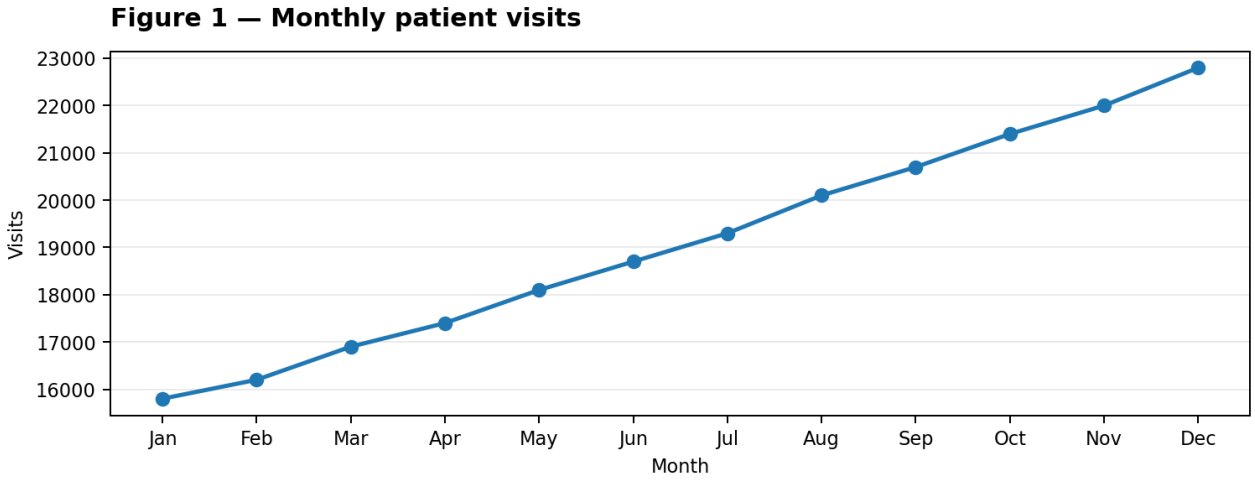
The sample covers a hypothetical January–December period. All figures are simulated and do not describe any real hospital, patient or clinical outcome. Department labels and metrics are used for analytical demonstration only.

Metric notes

Waiting time is shown in minutes; occupancy and readmission are shown as percentages. In real hospital reporting, metric definitions, observation windows, case mix and data quality rules should be agreed before comparing departments. Patient privacy and applicable data-protection requirements must be followed.

SECTION I · PATIENT VOLUME & DEPARTMENT ACTIVITY

Figure 1 — Monthly patient visits



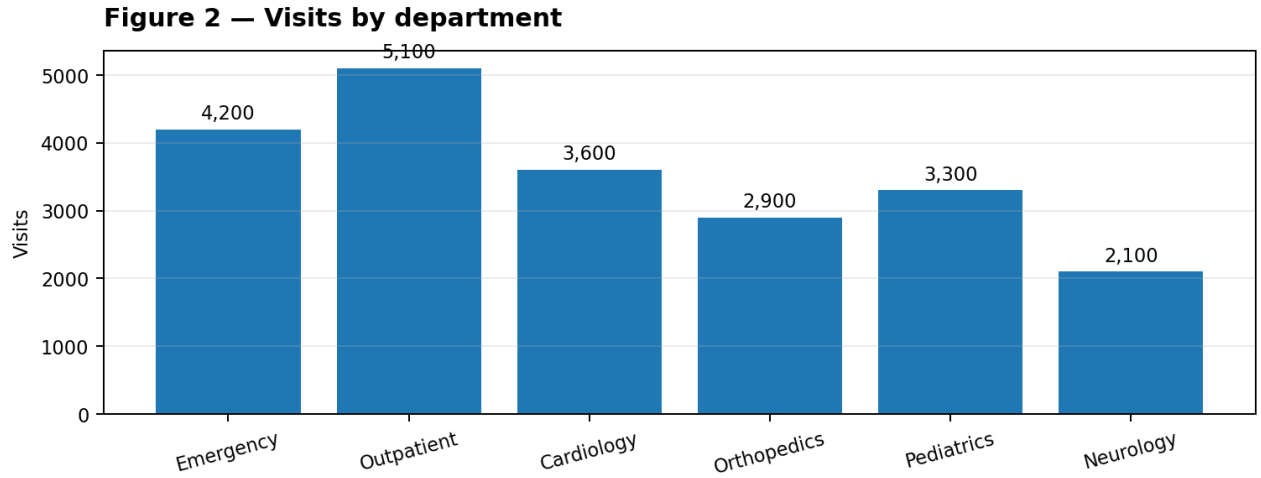
Note: All figures are simulated for educational demonstration; they are not actual patient or hospital records.

Month	Patient visits
Jan	15,800
Feb	16,200
Mar	16,900
Apr	17,400
May	18,100
Jun	18,700
Jul	19,300
Aug	20,100
Sep	20,700
Oct	21,400
Nov	22,000
Dec	22,800

Interpretation: The simulated series increases from 15,800 visits in January to 22,800 in December. This could prompt a real hospital to review staffing and capacity, but the sample trend is not a forecast.

Department activity

Figure 2 — Visits by department



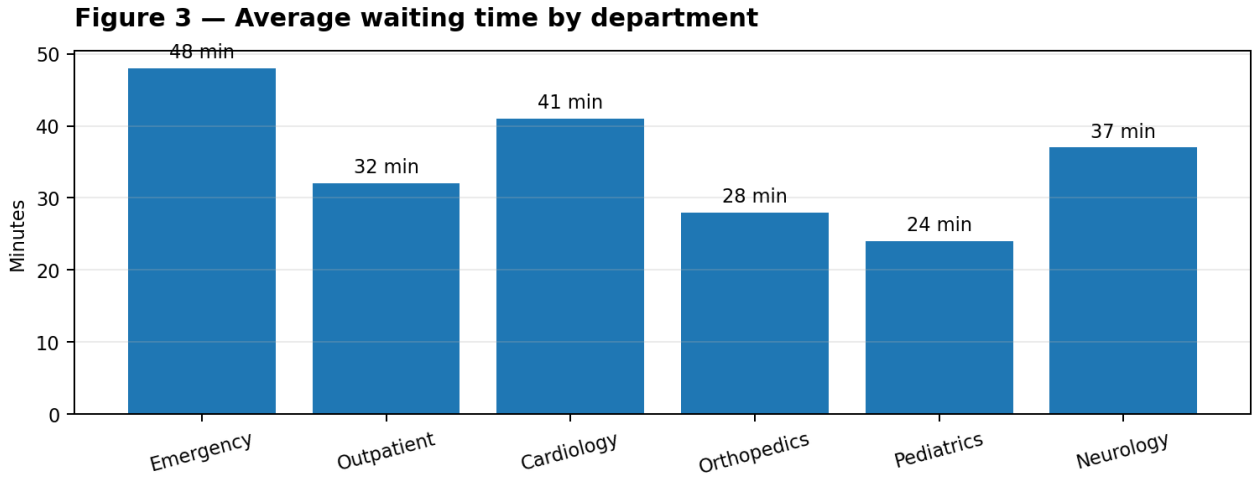
Note: All figures are simulated for educational demonstration; they are not actual patient or hospital records.

Rank	Department	Visits
1	Outpatient	5,100
2	Emergency	4,200
3	Cardiology	3,600
4	Pediatrics	3,300
5	Orthopedics	2,900
6	Neurology	2,100

Interpretation: Outpatient records the highest visit count in this sample, followed by Emergency. High volume alone does not indicate quality or urgency; it should be reviewed alongside acuity, staffing and outcomes.

Service efficiency

Figure 3 — Average waiting time by department



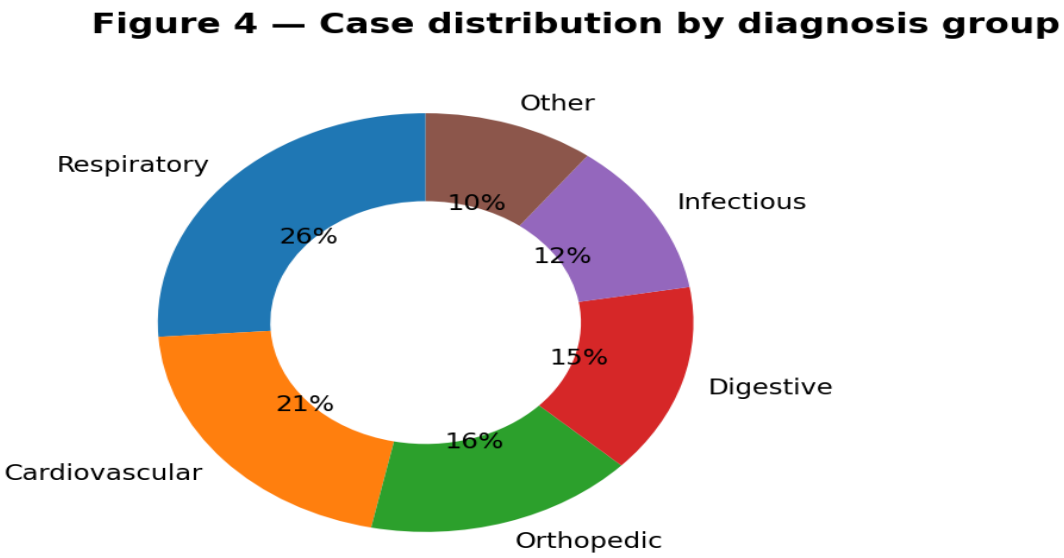
Note: All figures are simulated for educational demonstration; they are not actual patient or hospital records.

Rank	Department	Average wait
1	Emergency	48 min
2	Cardiology	41 min
3	Neurology	37 min
4	Outpatient	32 min
5	Orthopedics	28 min
6	Pediatrics	24 min

Interpretation: Emergency has the longest illustrative average wait. In practice, waiting times should be segmented by triage severity, arrival time and service type before drawing conclusions.

SECTION II · CASE MIX, DEMOGRAPHICS & ADMISSIONS

Figure 4 — Case distribution by diagnosis group



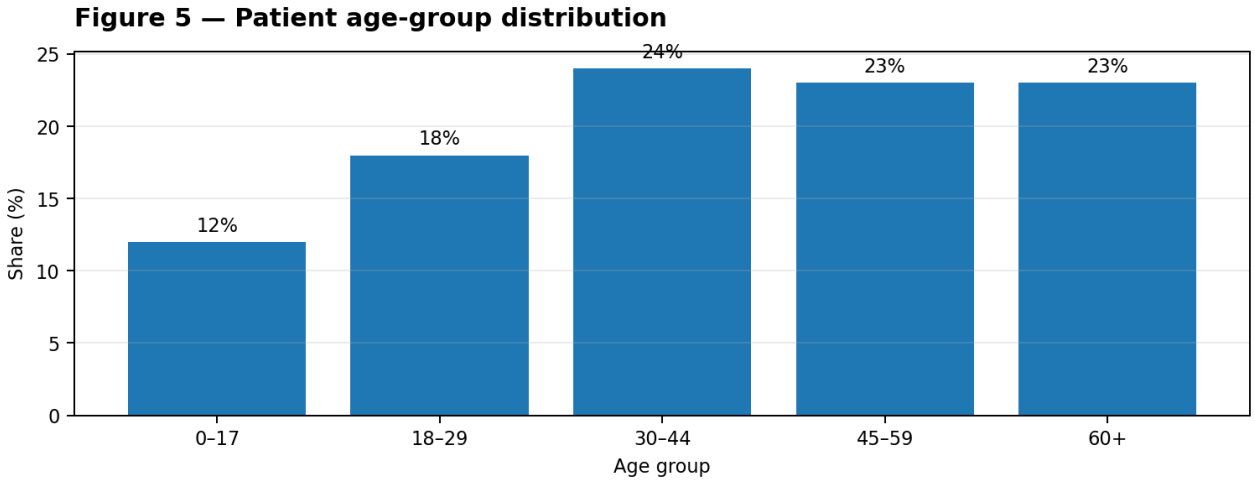
Note: All figures are simulated for educational demonstration; they are not actual patient or hospital records.

Rank	Diagnosis group	Cases
1	Respiratory	2,400
2	Cardiovascular	1,900
3	Orthopedic	1,500
4	Digestive	1,350
5	Infectious	1,100
6	Other	950

Interpretation: Respiratory cases are the largest group in this simulated dataset. This is a descriptive category breakdown only and does not indicate disease prevalence in any real population.

Patient demographics

Figure 5 — Patient age-group distribution



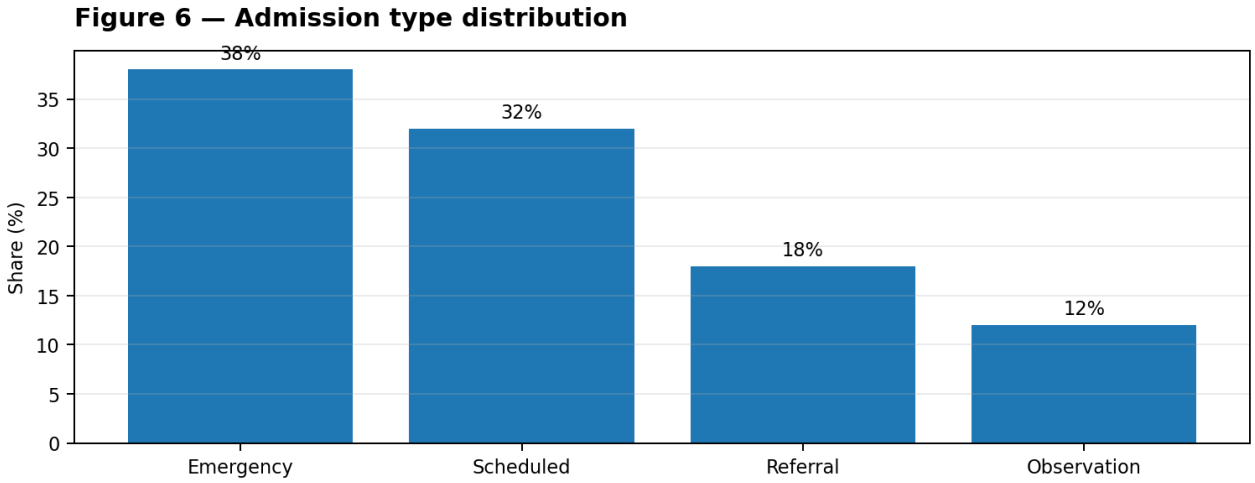
Note: All figures are simulated for educational demonstration; they are not actual patient or hospital records.

Rank	Age group	Share
1	30–44	24%
2	45–59	23%
3	60+	23%
4	18–29	18%
5	0–17	12%

Interpretation: Adults aged 30 and above make up most of the illustrative distribution. A real analysis should use only the minimum necessary demographic detail and protect patient confidentiality.

Admission patterns

Figure 6 — Admission type distribution



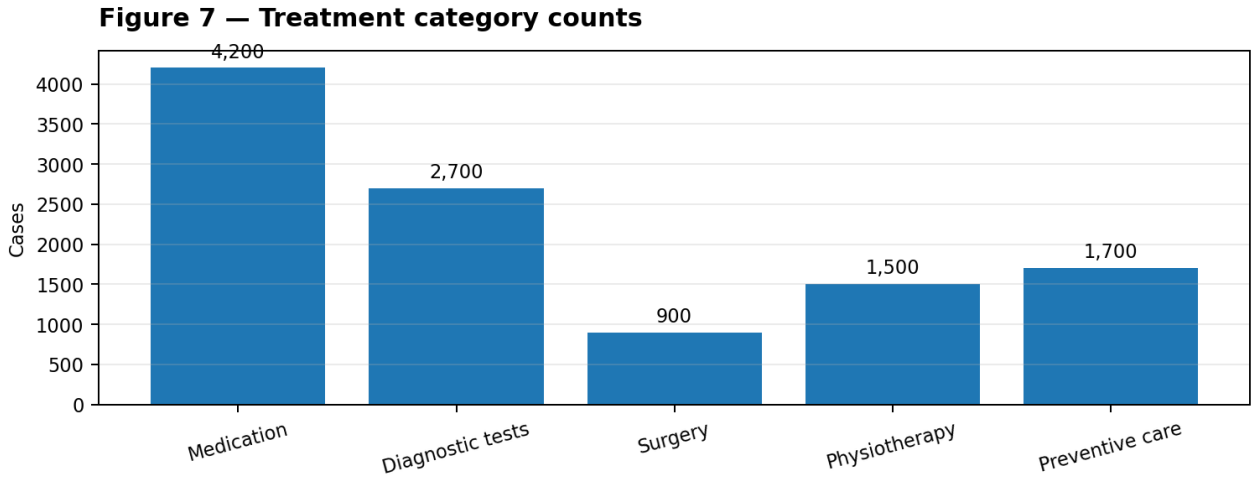
Note: All figures are simulated for educational demonstration; they are not actual patient or hospital records.

Rank	Admission type	Share
1	Emergency	38%
2	Scheduled	32%
3	Referral	18%
4	Observation	12%

Interpretation: Emergency admissions form the largest share in this sample. Operational planning should account for unpredictable arrivals and should not interpret these percentages as a measure of hospital performance by themselves.

SECTION III · TREATMENT, CAPACITY & READMISSIONS

Figure 7 — Treatment category counts



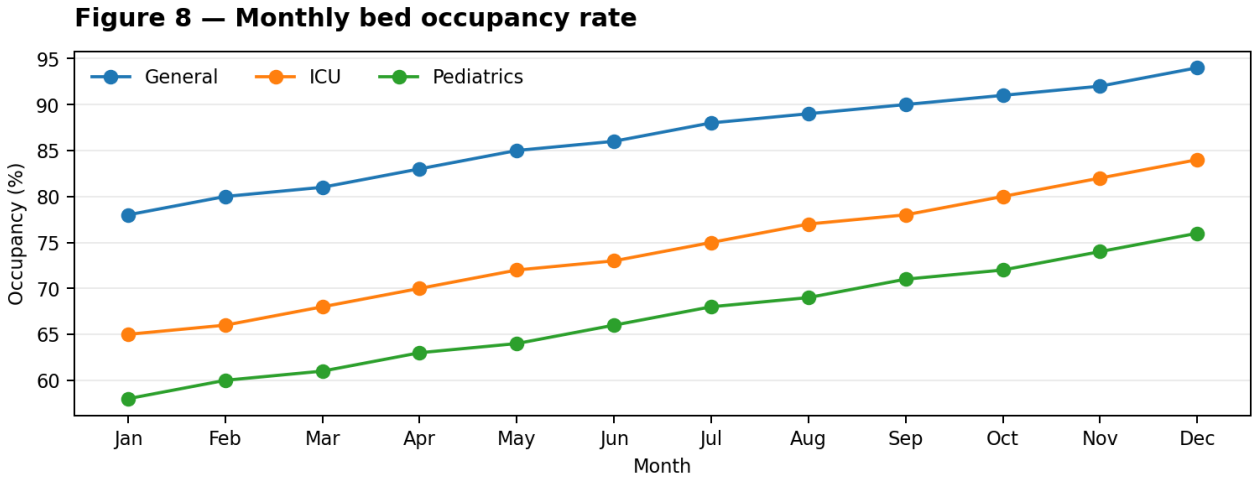
Note: All figures are simulated for educational demonstration; they are not actual patient or hospital records.

Rank	Treatment category	Cases
1	Medication	4,200
2	Diagnostic tests	2,700
3	Preventive care	1,700
4	Physiotherapy	1,500
5	Surgery	900

Interpretation: Medication is the largest category in this illustrative dataset. Treatment counts can overlap if a patient receives multiple services, so totals should not automatically be treated as unique patients.

Capacity utilization

Figure 8 — Monthly bed occupancy rate



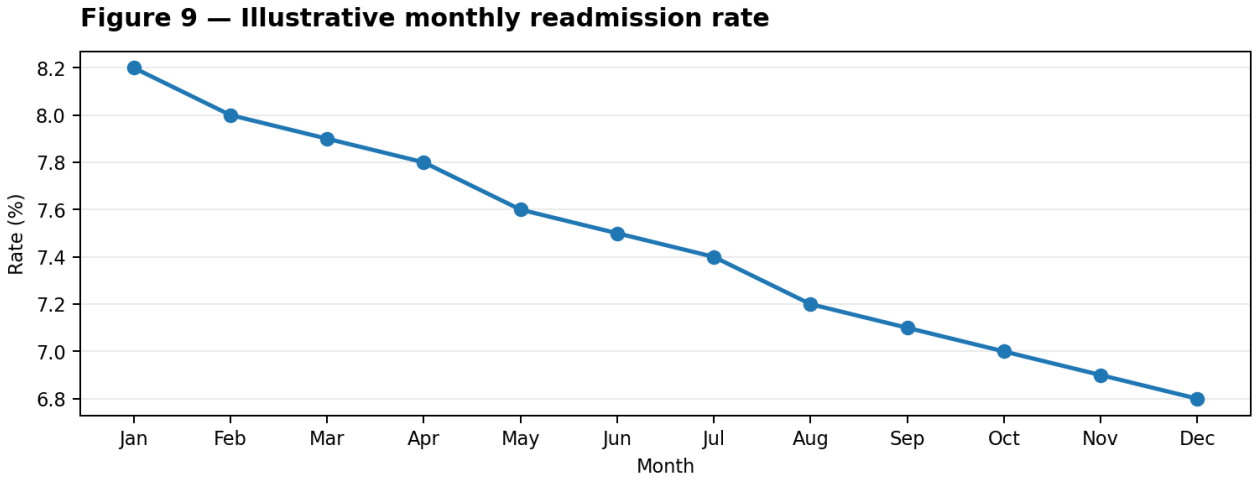
Note: All figures are simulated for educational demonstration; they are not actual patient or hospital records.

Ward type	January	December	Change
General	78%	94%	+16 pp
ICU	65%	84%	+19 pp
Pediatrics	58%	76%	+18 pp

Interpretation: Occupancy rises in all three hypothetical ward groups. High occupancy can constrain flexibility, but appropriate target ranges depend on local context and should be reviewed with clinical and operational leaders.

Care continuity indicator

Figure 9 — Illustrative monthly readmission rate



Note: All figures are simulated for educational demonstration; they are not actual patient or hospital records.

Month	Readmission rate
January	8.2%
April	7.8%
July	7.4%
October	7.0%
December	6.8%

Interpretation: The simulated rate trends downward. Readmission comparisons in real healthcare require a defined time window, eligibility rules, risk adjustment and careful interpretation; a lower rate alone does not prove better care.

SECTION IV · SUMMARY, FINDINGS & CONCLUSION

Figure 10 — Healthcare performance snapshot

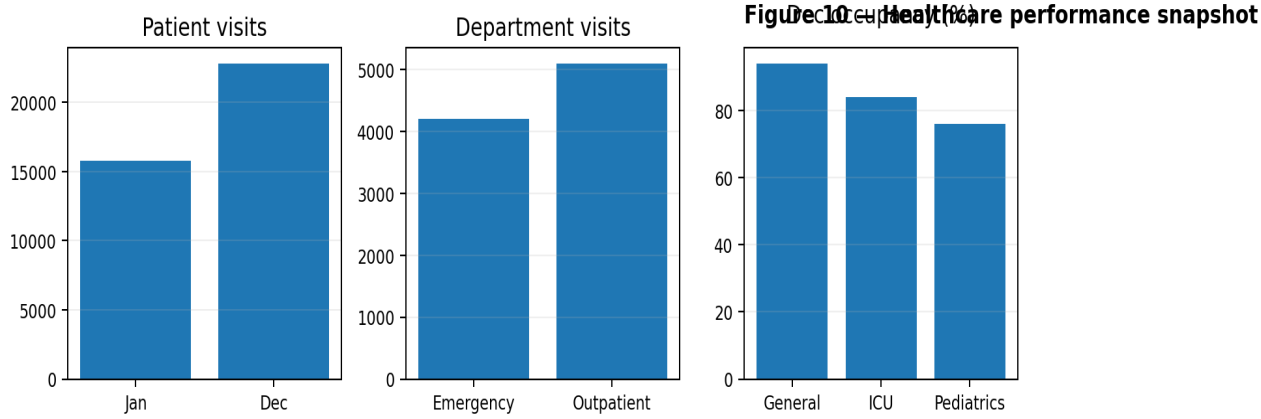


Figure 10 — Healthcare performance snapshot

Note: All figures are simulated for educational demonstration; they are not actual patient or hospital records.

Key metric	Illustrative result
Patient visits (January)	15,800
Patient visits (December)	22,800
Highest-volume department	Outpatient — 5,100 visits
Longest average wait	Emergency — 48 minutes
Largest diagnosis group	Respiratory — 2,400 cases
December general ward occupancy	94%
December ICU occupancy	84%
December readmission rate	6.8%

Key findings

1. Monthly visits rise throughout the simulated year.
2. Outpatient has the highest visit volume among the listed departments.
3. Emergency has the longest average waiting time in the sample.
4. General ward occupancy reaches 94% in December in the simulated series.
5. The readmission indicator declines in the illustrative dataset, but no clinical inference can be made from fictional data.

SECTION V · RECOMMENDATIONS & DATA NOTES

Recommendations

- Monitor patient volumes by day, time and department to support staffing decisions.
- Review waiting time with acuity, triage category and service type.
- Track bed occupancy alongside available staffed beds, transfers and discharge patterns.
- Define readmission measures and use risk-adjusted comparisons where appropriate.
- Validate data quality, missing values, duplicates and metric definitions before publishing dashboards.
- Use de-identified or aggregated data and follow applicable privacy and security requirements.

Conclusion

Healthcare data analytics can help organize operational indicators into clear visual summaries. This report demonstrates a structure for reviewing visits, workload, waiting time, case mix, admissions, treatment categories, bed occupancy and readmission patterns. Because all data is simulated, the findings are illustrative and should not be used to make clinical or staffing decisions.

Data accuracy and limitations

Every number in this report is fictional sample data created for educational visualization practice. The report does not use real patient records, hospital exports or verified clinical datasets. Diagnosis groups, demographics, bed occupancy and readmission values are illustrative only.

End of report · Healthcare Data Analytics · October 2026